

SECTION K IMPLICIT DIFFERENTIATION

You don't have to simplify your answers.

1. $-\frac{x}{y}$
2. $\frac{6y - 3x^2}{-6x - 3y^2}$
3. $\frac{1 - 3y^3 - 2xy}{x^2 + 9xy^2}$
4. $\frac{-1 + 10xy - 3x^2y^2}{2x^3y - 5x^2}$
5. $\frac{2x(x - y)^2 - (x - y) + (x + y)}{(x - y) + (x + y)} = \frac{x^3 - 2x^2y + xy^2 + y}{x}$
6. $\frac{\frac{y}{2\sqrt{xy}}}{1 - \frac{x}{2\sqrt{xy}}} = \frac{y}{2\sqrt{xy} - x}$
7. $\frac{1 - (35(x^2 + 3y^2)^{34})2x}{(35(x^2 + 3y^2)^{34})6y}$
8. $\frac{-y \sin(xy)}{1 + x \sin(xy)}$
9. $\frac{1 - (\cos(x^2y^2))2xy^2}{(\cos(x^2y^2))2x^2y}$
10. $\frac{1 - [3(\tan^2(xy^2 + y))(\sec^2(xy^2 + y))]y^2}{[3(\tan^2(xy^2 + y))(\sec^2(xy^2 + y))](2xy + 1)}$
11. $\frac{-y}{x}$